

Microbial community analysis

Workflow has been built with help of “Orchestrating microbiome analysis with R and Bioconductor” (Leo Lahti et. al), various analysis tool guides and own modifications.

Define file locations

```
# Path variables
asvfile <- "result_tables/asvs.tsv"
metafile <- "result_tables/metadata.tsv"
taxafile <- "result_tables/taxonomy.tsv"
treefile <- "result_tables/tree.nwk"
```

Import data

Data import and creation of a TreeSummarizedExperiment object.

```
# Abundance data is imported from txt file, rownames stored and emptied
counts <- read_tsv(asvfile, show_col_types = F)
ASV_names <- counts$Variant
counts$Variant <- NULL
# Metadata is imported from txt file, rownames stored and emptied
samples <- read_tsv(metafile, show_col_types = F)
sampleid <- samples$sampleid
samples$sampleid <- NULL
# Taxonomy table is imported txt file, rownames stored and emptied
taxonomy <- read_tsv(taxafile, show_col_types = F)
taxanames <- taxonomy$Variant
taxonomy$Variant <- NULL
# Check if there are discrepancies between data tables
if( any( colnames(counts) != sampleid ) ){
  counts <- counts[ , sampleid ]}
if( any( ASV_names != taxanames ) ){
  counts <- counts[ taxanames, ]}
# Numeric matrix format is required for abundance data
counts <- as.matrix(counts)
# It should be added to a SimpleList
assays <- SimpleList(counts = counts)
# colData and rowData should be in DataFrame (S4) format
samples <- DataFrame(samples)
taxonomy <- DataFrame(taxonomy)
# Create a TreeSummarizedExperiment object
tse <- TreeSummarizedExperiment(assays = assays,
                                colData = samples,
                                rowData = taxonomy)
# Add variant names as rownames
rownames(tse) <- ASV_names
```

Add phylogenetic tree to object

```
# Phylogenetic tree in newick format
phytree <- read.tree(treefile)
rowTree(tse) <- phytree
# Save as a rds file
saveRDS(tse,"rds/tse.rds")
```

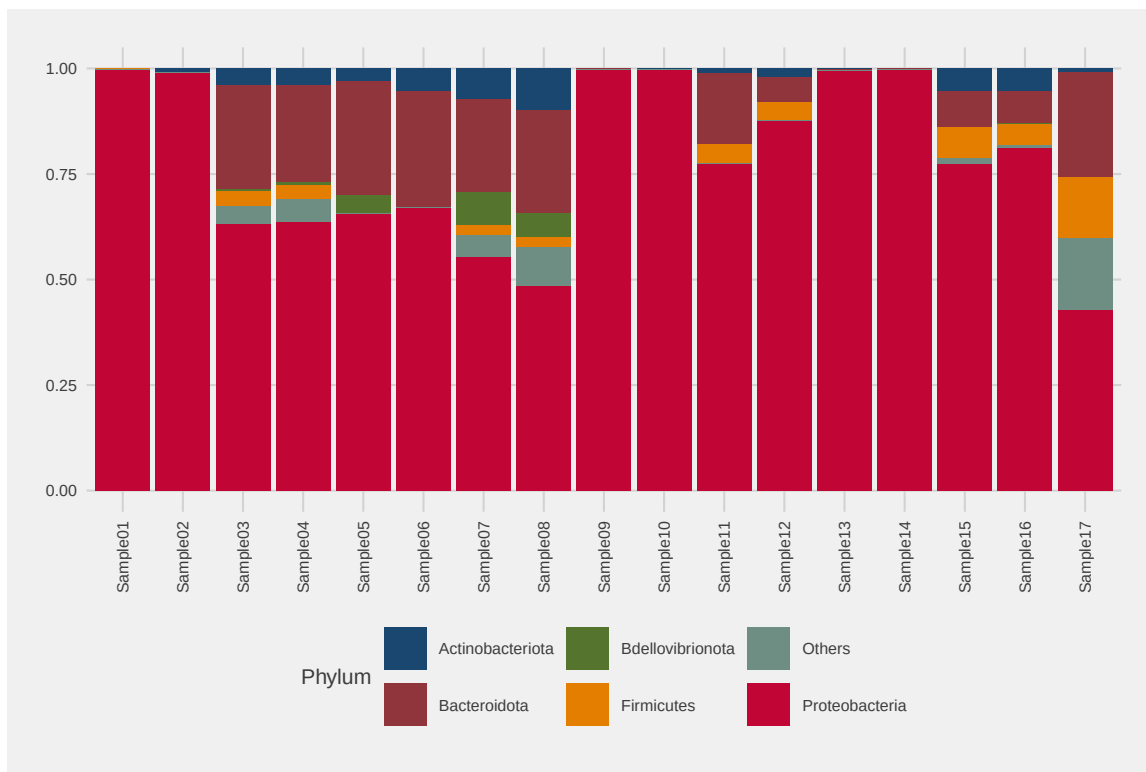
Community composition

Community composition at different taxonomic ranks can be studied by agglomerating taxa and using `getTop` function. Easiest way to visualize most common taxonomic features are barplots

Phylum rank

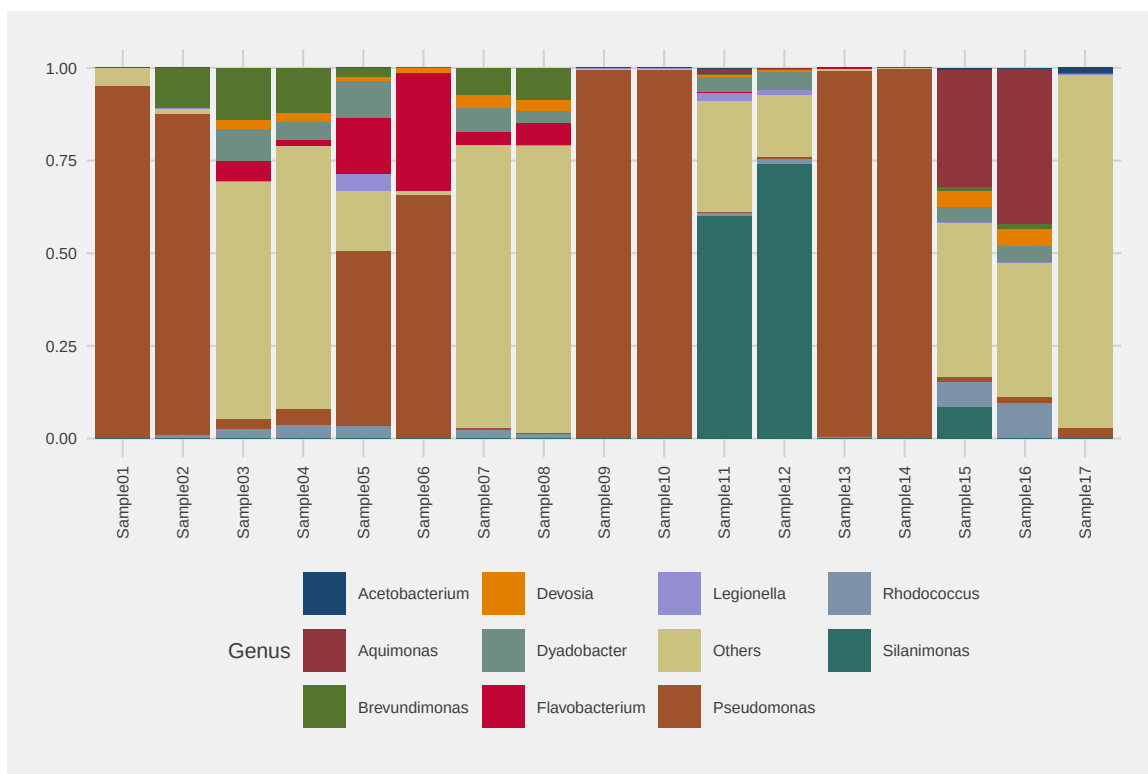
```
# Reload data
tse <- readRDS("rds/tse.rds")
n <- 5
# Agglomerate to genus rank
tse <- agglomerateByRank(tse, rank="Phylum", na.rm=T)
# Calculate relative abundance
tse <- transformAssay(tse, method="relabundance")
# Get top 10 features
top_features <- getTop(tse, top=n, method="median",
  assay.type="relabundance")
# Get data and filter using top features
df_phylum <- data.frame(assay(tse, "relabundance"))
# Proportion of Others bacteria group is 1 - colsums
df_phylum <- df_phylum %>% rownames_to_column(var="Phylum") %>%
  filter(Phylum %in% top_features) %>%
  bind_rows(summarise(., Phylum = "Others", across(where(is.numeric),
    ~ 1 - sum(.))))
# Create long table for ggplot
phylum_long <- pivot_longer(df_phylum, names_to="Sample",
  values_to="Abundance", cols = starts_with("Sample"))

# Barplot settings
bar_theme <- list(theme_fivethirtyeight(base_size=8),
  scale_fill_stata(), theme(axis.text.x =
  element_text(angle = 90,
  vjust = 0.5,
  hjust = 1)))
# Barplot
barplot1 <- ggplot(phylum_long, aes(x = Sample,
  y = Abundance, fill = Phylum)) +
  geom_bar(stat = "identity") + xlab("Sample") +
  bar_theme
barplot1
```



Genus rank

```
# Reload data
tse <- readRDS("rds/tse.rds")
n <- 10
# Agglomerate to genus rank
tse <- agglomerateByRank(tse, rank="Genus", na.rm=T)
# Calculate relative abundance
tse <- transformAssay(tse, method="relabundance")
# Get top 10 features
top_features <- getTop(tse, top=n, method="median",
  assay.type="relabundance")
# Get data and filter using top features
df_genus <- data.frame(assay(tse, "relabundance"))
# Proportion of Others bacteria group is 1 - colsums
df_genus <- df_genus %>% rownames_to_column(var="Genus") %>%
  filter(Genus %in% top_features) %>%
  bind_rows(summarise(., Genus = "Others", across(where(is.numeric),
    ~ 1 - sum(.))))
# Create long table for ggplot
genus_long <- pivot_longer(df_genus, names_to="Sample",
  values_to="Abundance", cols = starts_with("Sample"))
# Barplot
barplot2 <- ggplot(genus_long, aes(x = Sample,
  y = Abundance, fill = Genus)) +
  geom_bar(stat = "identity") + xlab("Sample") +
  bar_theme
barplot2
```



Abundance information can also be arranged to tables. Samples are merged by categorical data, taxonomy is agglomerated and data arranged by abundance.

```
# Reload tse
tse <- readRDS("rds/tse.rds")
#Merge filtered values, recount relative abundance and agglomeration to Genus
col4 <- mergeCols(tse,colData(tse)$Filtered)
col4 <- agglomerateByRank(col4, "Genus")
col4 <- relAbundanceCounts(col4)
#Create data frames for merged groups
opt1 <- data.frame(assay(col4,"relabundance")) %>%
  rownames_to_column(var = "opt1_asv") %>%
  arrange(desc(no)) %>% dplyr::select('Non-filtered' = opt1_asv,
    'Rel Abundance' = no)
opt2 <- data.frame(assay(col4,"relabundance")) %>%
  rownames_to_column(var = "opt2_asv") %>%
  arrange(desc(yes)) %>% dplyr::select(Filtered = opt2_asv,
    'Rel Abundance' = yes)
#How many to list in table
n <- 10
col4_table <- cbind(opt1[1:n,], opt2[1:n,])
kable(col4_table, digits = 2, caption = "Common taxa by Filtering") %>%
  kable_styling(latex_options = c("HOLD_position","striped"), font_size = 12) %>%
  row_spec(0, background = "indigo", color = "ivory")
```

Table 1: Common taxa by Filtering

Non-filtered	Rel Abundance	Filtered	Rel Abundance
Silanimonas	0.15	Pseudomonas	0.76
Brevundimonas	0.07	Flavobacterium	0.12
Pedobacter	0.05	Brevundimonas	0.03
Dyadobacter	0.05	Dyadobacter	0.02
Algoriphagus	0.05	Chryseobacterium	0.02
Aquimonas	0.04	Peredibacter	0.01
Alishewanella	0.04	Legionella	0.01
Kaistia	0.04	Rhodococcus	0.01
Bacteriovorax	0.03	Devosia	0.01
Flavobacterium	0.03	Burkholderia-Caballeronia-Paraburkholderia	0.00

Algae groups

```
#Merge filtered values, recount relative abundance and agglomeration to Genus
col5 <- mergeCols(tse, colData(tse)$Algae)
col5 <- agglomerateByRank(col5, "Genus")
col5 <- relAbundanceCounts(col5)
#Create data frames for merged groups
opt1 <- data.frame(assay(col5,"relabundance")) %>%
  rownames_to_column(var = "opt1_asv") %>%
  arrange(desc(chlorella_s)) %>% dplyr::select(Chlorella = opt1_asv,
  'Rel Abundance' = chlorella_s)
opt2 <- data.frame(assay(col5,"relabundance")) %>%
  rownames_to_column(var = "opt2_asv") %>%
  arrange(desc(selenastrum)) %>% dplyr::select(Selenastrum = opt2_asv,
  'Rel Abundance' = selenastrum)
#How many to list in table
n <- 10
col5_table <- cbind(opt1[1:n,], opt2[1:n,])
kable(col5_table, digits = 2, caption = "Common taxa by Algae") %>%
  kable_styling(latex_options = c("HOLD_position", "striped"), font_size = 12) %>%
  row_spec(0, background = "indigo", color = "ivory")
```

Table 2: Common taxa by Algae

Chlorella	Rel Abundance	Selenastrum	Rel Abundance
Pseudomonas	0.59	Pseudomonas	0.41
Silanimonas	0.13	Flavobacterium	0.14
Brevundimonas	0.06	Dyadobacter	0.04
Algoriphagus	0.04	Chryseobacterium	0.03
Alishewanella	0.03	Brevundimonas	0.03
Dyadobacter	0.02	Aquimonas	0.03
Rhodococcus	0.01	Pedobacter	0.03
Pedobacter	0.01	Kaistia	0.02
Prostheco bacter	0.01	Bacteriovorax	0.02
Flavobacterium	0.01	Devosia	0.02

Age groups

```
#Merge filtered values, recount relative abundance and agglomeration to Genus
col3 <- mergeCols(tse, as.character(colData(tse)$Age))
col3 <- agglomerateByRank(col3, "Genus")
col3 <- relAbundanceCounts(col3)
#Create data frames for merged groups
opt1 <- data.frame(assay(col3, "relabundance")) %>%
  rownames_to_column(var = "opt1_asv") %>%
  arrange(desc(X5)) %>% dplyr::select(Day5 = opt1_asv, 'Rel Abundance' = X5)
opt2 <- data.frame(assay(col3, "relabundance")) %>%
  rownames_to_column(var = "opt2_asv") %>%
  arrange(desc(X30)) %>% dplyr::select(Day30 = opt2_asv, 'Rel Abundance' = X30)
#How many to list in table
n <- 10
col3_table <- cbind(opt1[1:n,], opt2[1:n,])
kable(col3_table, digits = 2, caption = "Common taxa by Age") %>%
  kable_styling(latex_options = c("HOLD_position", "striped"), font_size = 12) %>%
  row_spec(0, background="indigo", color="ivory")
```

Table 3: Common taxa by Age

Day5	Rel Abundance	Day30	Rel Abundance
Pseudomonas	0.48	Pseudomonas	0.52
Flavobacterium	0.11	Silanimonas	0.22
Brevundimonas	0.06	Aquimonas	0.06
Dyadobacter	0.04	Algoriphagus	0.05
Chryseobacterium	0.03	Roseococcus	0.03
Pedobacter	0.03	Dyadobacter	0.02
Kaistia	0.02	Rhodococcus	0.02
Alishewanella	0.02	Roseomonas	0.01
Bacteriovorax	0.02	Acinetobacter	0.01
Devosia	0.01	Devosia	0.01

Alpha diversity

Diversity can be studied using diversity indexes. Values can be stored to **colData** under defined names. We create table with Shannon, Faith and observed features.

```
# Reload tse
tse <- readRDS("rds/tse.rds")
#Calculate Shannon index
tse <- addAlpha(tse, assay.type = "counts",
               index = "shannon_diversity",
               name = "Shannon")
#Calculate phylogenetic Faith index
tse <- addAlpha(tse, assay.type = "counts",
               index = "faith_diversity",
               name = "Faith", tree_name = "phylo")
#Calculate richness with Chao1 index
tse <- addAlpha(tse, assay.type = "counts",
               index = "observed_richness",
               name = "Observed")
#Create table
kable(data.frame(Sample = rownames(colData(tse)), Shannon = colData(tse)$Shannon,
               Faith = colData(tse)$Faith,
               Observed_features = colData(tse)$Observed), digits = 2) %>%
  kable_styling(latex_options = c("HOLD_position", "striped"),
               font_size = 10) %>%
  row_spec(0, background = "indigo", color = "ivory")
```

Sample	Shannon	Faith	Observed_features
Sample01	0.79	4.96	40
Sample02	1.43	4.96	51
Sample03	4.64	16.11	353
Sample04	4.58	16.86	362
Sample05	2.67	11.33	65
Sample06	1.68	12.12	52
Sample07	5.34	34.92	718
Sample08	5.24	34.36	690
Sample09	0.12	4.43	33
Sample10	0.12	3.59	29
Sample11	2.03	6.38	92
Sample12	1.95	7.69	102
Sample13	0.15	3.91	29
Sample14	0.10	3.88	31
Sample15	3.51	18.13	194
Sample16	3.38	17.47	164
Sample17	3.92	24.19	411

Boxplots can be used to compare diversity between categories

```
# Boxplot function
plot_shannon <- function(x_var, fill_var, title, x_label, scale_x = FALSE, s_var) {
  p <- ggplot(as.data.frame(colData(tse)), aes(x = as.character(.data[[x_var]]), y = Shannon, fill = .data[[fill_var]])) +
    geom_boxplot(alpha = 0.7, show.legend = FALSE) +
    labs(title = title, y = "Shannon", x = x_label) +
    theme_hc(base_size = s_var) + scale_fill_stata()

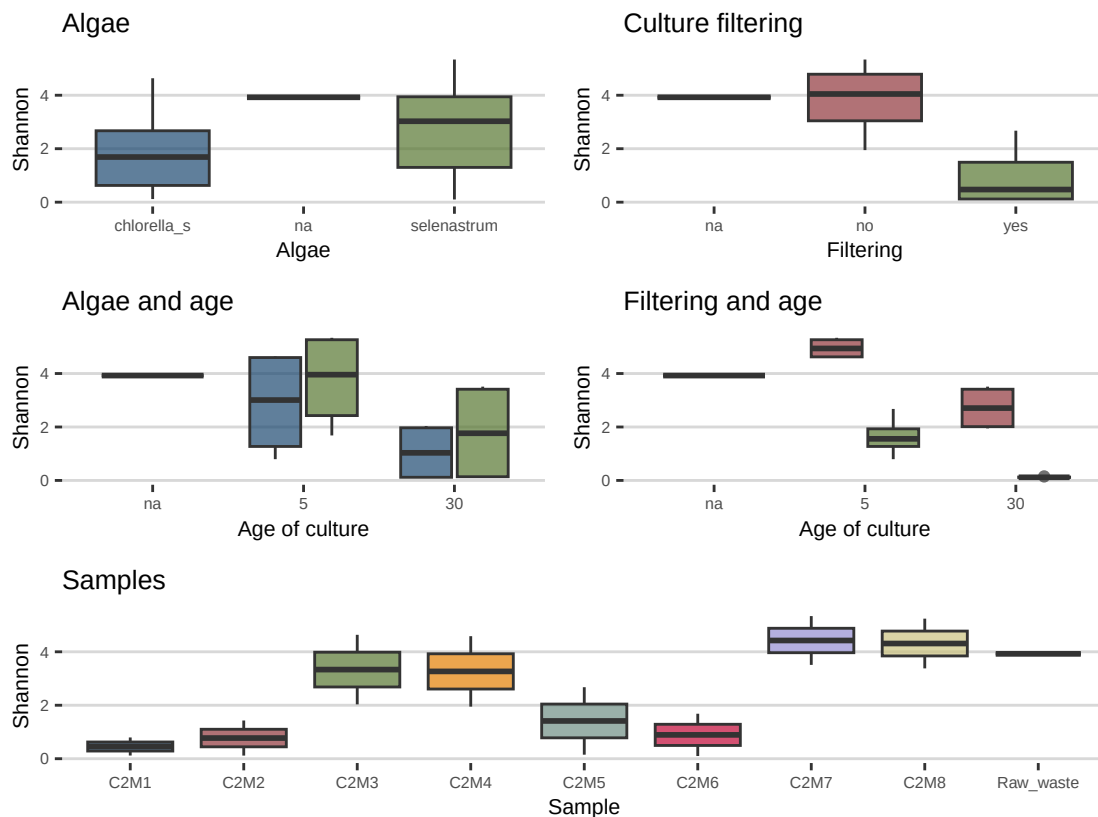
  if (scale_x) p <- p + scale_x_discrete(limits = rev)

  return(p)
}

# Generate plots
plot1 <- plot_shannon("Algae", "Algae", "Algae", "Algae", 8)
plot2 <- plot_shannon("Filtered", "Filtered", "Culture filtering",
  "Filtering", 8)
plot3 <- plot_shannon("Name", "Name", "Samples", "Sample", 8)
plot4 <- plot_shannon("Age", "Algae", "Algae and age", "Age of culture",
  scale_x = TRUE, 8)
plot5 <- plot_shannon("Age", "Filtered", "Filtering and age", "Age of culture",
  scale_x = TRUE, 8)
```

Alpha diversity boxplots

```
# Create the combined plot
combined_plot <- (plot1 | plot2) /
  (plot4 | plot5) /
  plot3
# Display the combined plot
combined_plot
```



Filtering has clear effect on richness. Both algae species seem also to decrease richness compared to untreated sample. However, there is only one control and decrease is not statistically significant. Culture age also decreases diversity. This is most evident in plots displaying filtering and age categories separately.

Beta diversity

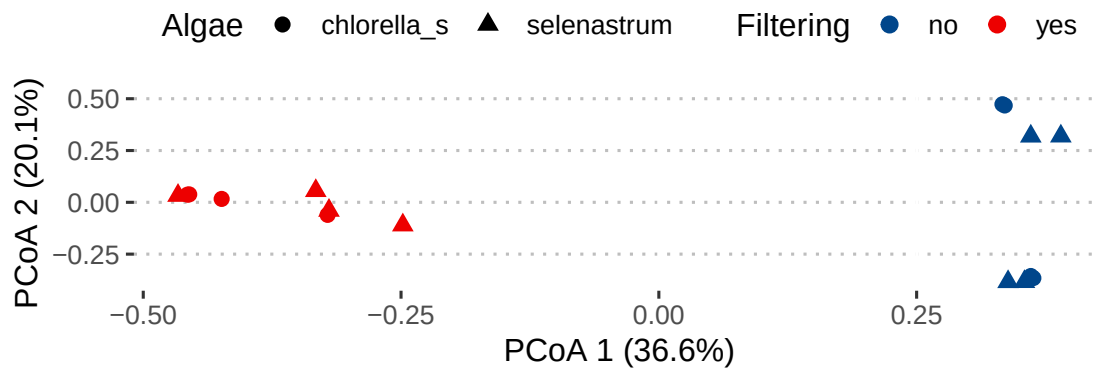
Bray-Curtis dissimilarity analysis using count data and rarefaction to 11 624.

```
# Reload data
tse <- readRDS("rds/tse.rds")
# Calculate Bray-Curtis distance and run MDS
tse <- runMDS(tse, FUN = getDissimilarity, method = "bray",
             assay.type = "counts", name = "MDS_bray",
             sample = 11624)
# Reduce to 2D
pcoa_bray <- plotReducedDim(tse, "MDS_bray")
# Calculate explained variance
e <- attr(reducedDim(tse, "MDS_bray"), "eig")
rel_eig <- e / sum(e[e > 0])
# Create dataframe for each axis and bind sample attributes
bray_attributes <- data.frame(pcoa1 = pcoa_bray$data[, 1],
                             pcoa2 = pcoa_bray$data[, 2],
                             Filtering = colData(tse)$Filtered,
                             Age = as.character(colData(tse)$Age),
                             Names = colData(tse)$Name,
                             Algae = colData(tse)$Algae[1:16,])
# Create series of plots using data frame
plot_pcoa <- function(data, color_var, shape_var, x_label, y_label) {
  ggplot(data, aes(x = pcoa1, y = pcoa2, color = !!sym(color_var),
                  shape = !!sym(shape_var))) + geom_point(size = 2.5) +
    theme_pubclean() + scale_color_lancet() + labs(x = x_label, y = y_label)
}
x_label <- paste("PCoA 1 (", round(100 * rel_eig[[1]], 1), "%)", sep = "")
y_label <- paste("PCoA 2 (", round(100 * rel_eig[[2]], 1), "%)", sep = "")
filtering_plot <- plot_pcoa(bray_attributes, "Filtering", "Algae",
                           x_label, y_label)
age_plot <- plot_pcoa(bray_attributes, "Age", "Algae", x_label, y_label)
```

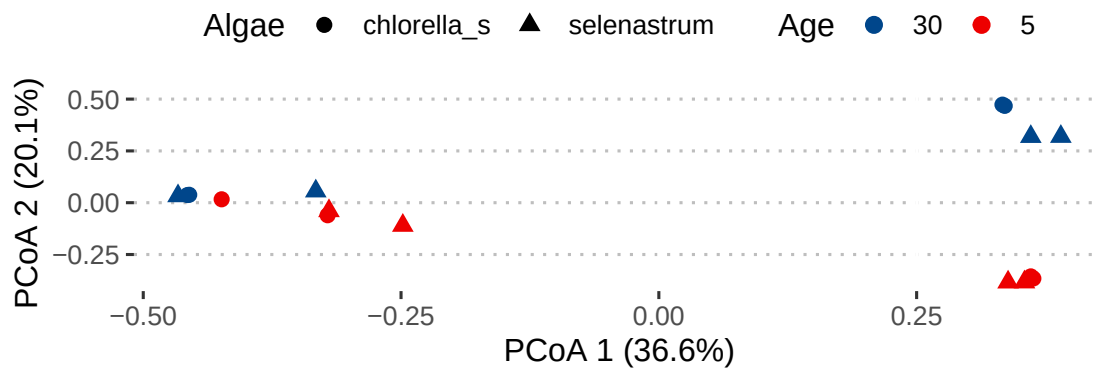
Plots

```
wrap_plots(filtering_plot + ggtitle("Filtering and Algae"), age_plot + ggtitle("Filtering and Age"), ncol=1)
```

Filtering and Algae



Filtering and Age



Filtering plays the most significant role in driving sample dissimilarity.

Beta diversity analysis using unifracs method

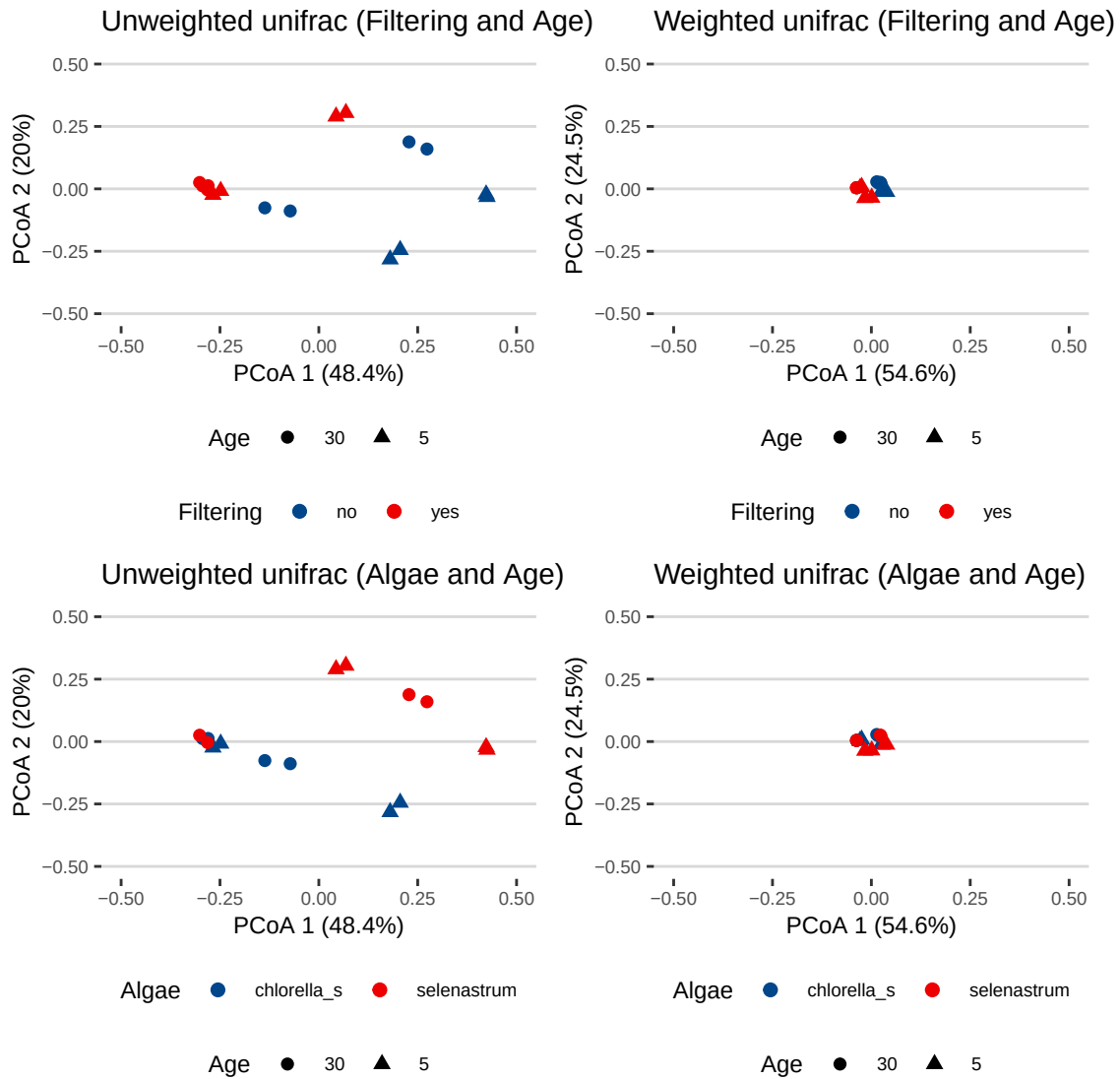
```
# Reload data
tse <- readRDS("rds/tse.rds")
# Method names for both unweighted and weighted UniFrac
methods <- c("unweighted_uni", "weighted_uni")
# Run MDS and create data frames in one loop
results <- lapply(methods, function(method) {
  weighted <- method == "weighted_uni"
  # Run MDS for each method
  tse <- runMDS(tse, FUN = getDissimilarity, method = "unifrac",
    name = method, assay.type = "counts", tree = rowTree(tse),
    sample = 11624, weighted = weighted)
  # Create PCoA data frame and calculate explained variance
  pcoa <- reducedDim(tse, method)
  pcoa_df <- data.frame(pcoa1 = pcoa[, 1],
    pcoa2 = pcoa[, 2],
    Filtering = colData(tse)$Filtered,
    Age = colData(tse)$Age,
    Algae = colData(tse)$Algae[1:16, ])
  # Calculate explained variance
  eig <- attr(pcoa, "eig")
  rel_eig <- eig / sum(eig[eig > 0])
  list(pcoa_df = pcoa_df, rel_eig = rel_eig)
})
# Extract results for each method
unweighted_df <- results[[1]]$pcoa_df
weighted_df <- results[[2]]$pcoa_df
rel_eig_unw <- results[[1]]$rel_eig
rel_eig_w <- results[[2]]$rel_eig
```

Next ggplot objects

```
# Define a function to create the plots
create_plot <- function(df, pcoa1, pcoa2, rel_eig, title, color_var, shape_var) {
  ggplot(data = df, aes(x = pcoa1, y = pcoa2, color = !!sym(color_var), shape = !!sym(shape_var))) + geom_point(size = 2) +
    labs(x = paste("PCoA 1 (", round(100 * rel_eig[1], 1), "%)", sep = " "),
    y = paste("PCoA 2 (", round(100 * rel_eig[2], 1), "%)", sep = " "),
    title = title) + scale_y_continuous(limits = c(-0.5, 0.5)) +
    scale_x_continuous(limits = c(-0.5, 0.5)) + theme_hc(base_size = 9) +
    scale_color_lancet() + theme(legend.position = "bottom",
    legend.box = "vertical",
    legend.margin = margin())
}
# List of plot parameters for looping
plots_params <- list(
  list(df = unweighted_df, pcoa1 = "pcoa1", pcoa2 = "pcoa2", rel_eig = rel_eig_unw,
    title = "Unweighted unifrac (Filtering and Age)", color_var = "Filtering", shape_var = "Age"),
  list(df = weighted_df, pcoa1 = "pcoa1", pcoa2 = "pcoa2", rel_eig = rel_eig_w,
    title = "Weighted unifrac (Filtering and Age)", color_var = "Filtering", shape_var = "Age"),
  list(df = unweighted_df, pcoa1 = "pcoa1", pcoa2 = "pcoa2", rel_eig = rel_eig_unw,
    title = "Unweighted unifrac (Algae and Age)", color_var = "Algae", shape_var = "Age"),
  list(df = weighted_df, pcoa1 = "pcoa1", pcoa2 = "pcoa2", rel_eig = rel_eig_w,
    title = "Weighted unifrac (Algae and Age)", color_var = "Algae", shape_var = "Age")
)
# Create the plots using the parameters
plots <- lapply(plots_params, function(params) {
  create_plot(params$df, params$pcoa1, params$pcoa2, params$rel_eig, params$title,
    params$color_var, params$shape_var)
})
```

Plot unifrac results

```
wrap_plots(plots[[1]], plots[[2]], plots[[3]], plots[[4]], ncol=2)
```



Results are similar to Bray-Curtis, but weighted plots show only small differences with filtering. Explained variance is high, though.

Permanova analysis

Permanova measures importance of each variable to total variance.

```
# Reload tse
tse <- readRDS("rds/tse.rds")
# Calculate relabundance
tse <- transformAssay(tse, method = "relabundance")
tse <- addRDA(tse, assay.type = "relabundance", sample=11624,
             formula = assay ~ Filtered + Age + Algae,
             distance = "bray")
rda_info <- attr(reducedDim(tse, "RDA"), "significance")
kable(rda_info$permanova, digits = 2) %>%
  kable_styling(latex_options = c("HOLD_position", "striped"),
               font_size = 12) %>%
  row_spec(0, background="indigo", color="ivory")
```

	Df	SumOfSqs	F	Pr(>F)	Total variance	Explained variance
Model	4	3.99	6.10	0.00	5.95	0.67
Filtered	1	2.25	13.76	0.00	5.95	0.38
Age	1	0.89	5.45	0.00	5.95	0.15
Algae	1	0.28	1.72	0.12	5.95	0.05
Residual	12	1.96	NA	NA	5.95	0.33

Variable groups explain 67 % of total variance. Filtering and Age are statistically significant, algae not.

We also need to check if the homogeneity assumption is fulfilled

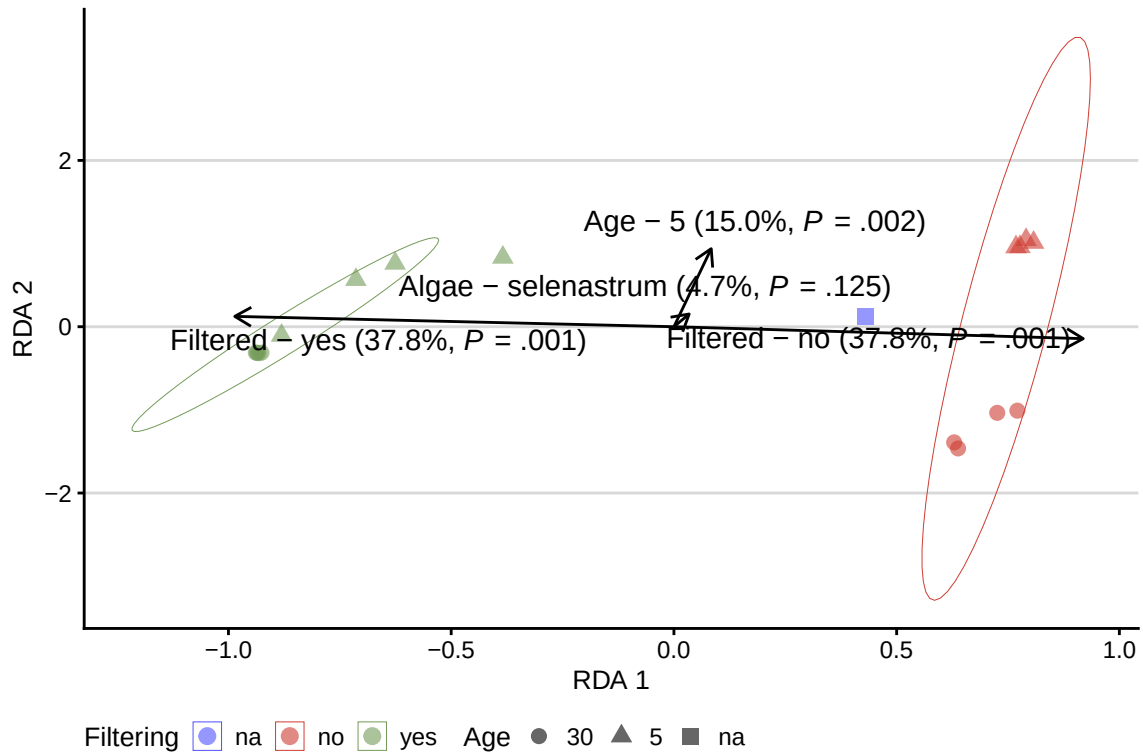
```
kable(rda_info$homogeneity, digits = 2) %>%
  kable_styling(latex_options = c("HOLD_position", "striped"),
               font_size = 12) %>%
  row_spec(0, background="indigo", color="ivory")
```

	Df	Sum Sq	Mean Sq	F	N.Perm	Pr(>F)	Total variance	Explained variance
Filtered	2	0.38	0.19	3.80	999	0.04	1.09	0.35
Age	2	0.25	0.13	0.96	999	0.58	2.09	0.12
Algae	2	0.29	0.15	7.19	999	0.01	0.58	0.51

Age does fulfill homogeneity, while filtering does fall slightly below 0.05 level. Algae groups are not homogenous. Results mean that in Filtered groups variance is quite different can contribute to differences measured.

The plotRDA function can be used to visualize results.

```
# Generate RDA plot
permanova <- plotRDA(tse, "RDA", colour_by="Filtered", shape="Age", add.ellipse="colour")
permanova$scales$scales <- NULL
#change dot size
permanova$layers[[1]]$aes_params$size <-2.5
#add theme and color palette
permanova + theme_hc(base_size = 10) +
  scale_colour_igv() + labs(colour="Filtering")
```



Differential abundance

Differential abundance can be studied with several R packages. Here we use ALDEx2, ANCOM-BC and Maaslin.

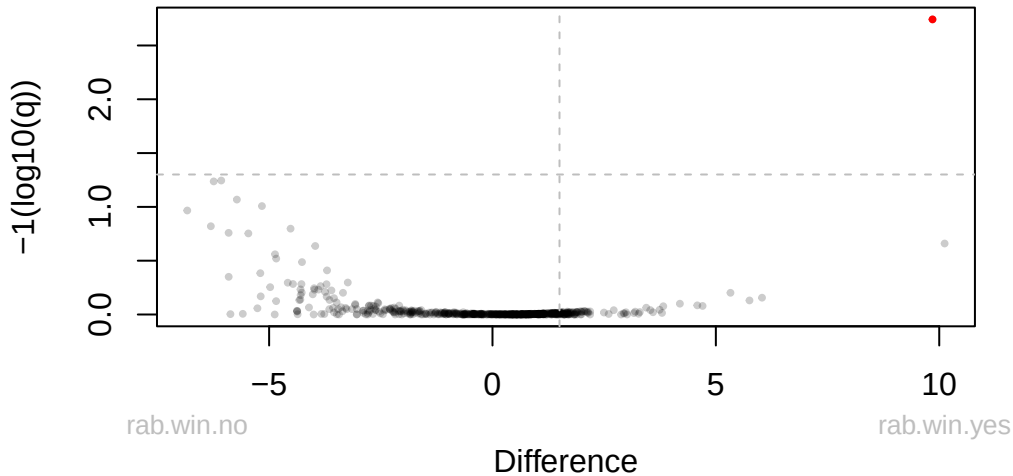
First, data is filtered by prevalence. Also taxonomy table is extracted for later use.

```
# Reload data
tse <- readRDS("rds/tse.rds")
#Filtering data based on prevalence. We also drop uncategorized raw wastewater
tse_daa <- subsetByPrevalentTaxa(tse[,1:16], detection = 0, prevalence = 0.1)
#We prepare also list of all taxa labels for later use
featureids <- as.data.frame(getTaxonomyLabels(tse_daa, make_unique=FALSE),
                             rownames(tse_daa))
featureids <- rownames_to_column(featureids, var="asv")
colnames(featureids) <- c("ASV", "taxon")
```

Aldex2 analysis on *filtered* category.

```
#ALDEx2 analysis can be performed in modular fashion
#aldex.clr - generates random instances of the centred log-ratio transformed values
filter_aldex <- aldex.clr(assay(tse_daa), tse_daa$Filtered, useMC = T, mc.samples=256, verbose = F)
#aldex.ttest - perform Welch's t and Wilcoxon test when there are only two conditions
filter_tt <- aldex.ttest(filter_aldex, paired.test = F, verbose = F)
#aldex.effect - estimate effect size and the within and between condition values
filter_effect <- aldex.effect(filter_aldex, CI = T, verbose = F)
#Merge two outputs
filter_aldex_out <- data.frame(filter_tt, filter_effect)
#Create plots
```

```
aldex.plot(filter_aldex_out, type="volcano", test="welch", xlab="Difference",
           ylab="-1(log10(q))", Main="Volcano plot")
```



In plot, red dots represent significantly changed taxa, grey dots are abundant taxa and black dots are rare taxa.

Aldex found five variants, where wilcoxon probability test result is $p \leq 0.05$.

```
#Filter significantly different taxa and create table
aldex_res <- rownames_to_column(filter_aldex_out, "genus")
aldex_res <- aldex_res %>% dplyr::filter(wi.eBH <= 0.05) %>%
  dplyr::select(genus, we.eBH, wi.eBH, effect, overlap)
#Merge genus id and taxa names into single table
identity <- merge(aldex_res, featureids, by.x=c("genus"), by.y=c("ASV"))
identity <- identity %>% relocate("genus", "taxon", "we.eBH", "wi.eBH",
                                "effect", "overlap")
kable(identity, digits=2) %>% kable_styling(latex_options = c("HOLD_position",
                                                            "striped"),
                                       font_size = 12) %>%
  row_spec(0, background="indigo", color="ivory")
```

genus	taxon	we.eBH	wi.eBH	effect	overlap
ASV29	Family:Carnobacteriaceae	0.09	0.01	-1.74	0.01
ASV64	Genus:Brevundimonas	0.06	0.00	-2.14	0.00
ASV70	Genus:Dyadobacter	0.06	0.00	-1.85	0.00
ASV78	Family:Comamonadaceae	0.18	0.03	-1.39	0.03
ASV86	Family:Devosiaceae	0.10	0.00	-1.85	0.00

Testing algae or culture age did not provide results.

Ancom-BC2 (Analysis of Compositions of Microbiomes with Bias Correction)

```
#Perform the analysis
ancom_out = ancombc2(data = tse_daa, assay_name = "counts",
  fix_formula = "Filtered + Algae + Age",
  p_adj_method = "holm", prv_cut = 0, lib_cut = 0,
  group = "Filtered", struc_zero = TRUE, global = TRUE)
saveRDS(ancom_out, "rds/ancom_out.rds")
```

```
ancom_out <- readRDS("rds/ancom_out.rds")
```

Results are collected into out\$res data frame. We can filter significant variants.

First variable is Filtering (yes/no)

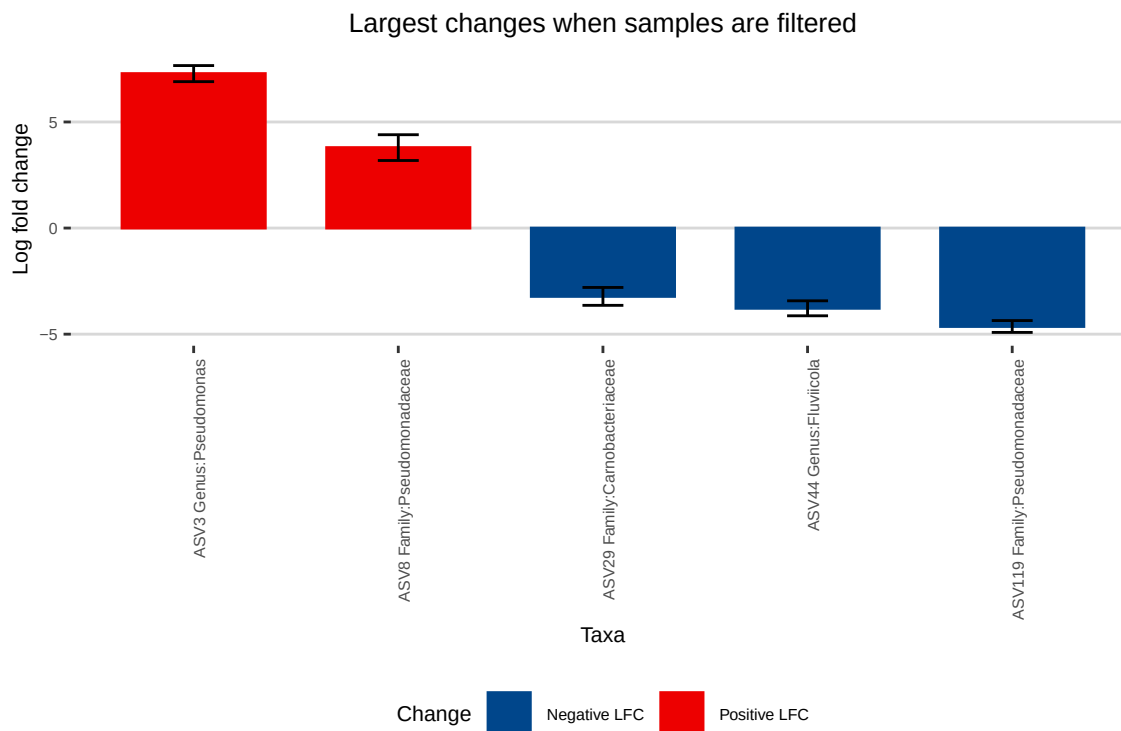
```
#Create data frame, filter Diff = TRUE and arrange by Lfc
#We also combine taxaid and taxonomic name into first column for our figure
df_filtered <- data.frame(ASV = ancom_out$res$taxon,
  Lfc = ancom_out$res$lfc_Filteredyes,
  SE = ancom_out$res$se_Filteredyes,
  Q = ancom_out$res$q_Filteredyes,
  Diff = ancom_out$res$diff_Filteredyes) %>%
  filter(Diff == "TRUE") %>% arrange(desc(Lfc)) %>%
  left_join(featureids, by = "ASV")
df_filtered$ASV <- paste(df_filtered$ASV, df_filtered$taxon)
df_filtered <- df_filtered %>% dplyr::select(, -6) %>%
  mutate(Change = ifelse(Lfc > 0, "Positive LFC", "Negative LFC"))
kable(df_filtered, caption="Taxa that are changed by filtering",
  digits=2,) %>% kable_styling(latex_options = c("HOLD_position",
  "striped"),
  font_size = 12) %>%
  row_spec(0, background="indigo", color="ivory")
```

Table 8: Taxa that are changed by filtering

ASV	Lfc	SE	Q	Diff	Change
ASV3 Genus:Pseudomonas	7.28	0.38	0.00	TRUE	Positive LFC
ASV8 Family:Pseudomonadaceae	3.79	0.61	0.03	TRUE	Positive LFC
ASV29 Family:Carnobacteriaceae	-3.22	0.42	0.01	TRUE	Negative LFC
ASV44 Genus:Fluviicola	-3.78	0.35	0.04	TRUE	Negative LFC
ASV119 Family:Pseudomonadaceae	-4.64	0.28	0.05	TRUE	Negative LFC

Bar plot of log fold changes including standard error

```
#Create ordered taxa list
p_filter <- ggplot(data = df_filtered,
  aes(x = factor(ASV, level=df_filtered$ASV), y = Lfc, fill = Change,
  color = Change)) + geom_bar(stat = "identity", width = 0.7,
  position = position_dodge(width = 0.4)) +
  geom_errorbar(aes(ymin = Lfc - SE, ymax = Lfc + SE), width = 0.2,
  position = position_dodge(0.05), color = "black") +
  labs(x = "Taxa", y = "Log fold change",
  title = "Largest changes when samples are filtered") +
  theme_hc(base_size = 8) + scale_fill_lancet() + scale_color_lancet() +
  theme(plot.title = element_text(hjust = 0.5),
  panel.grid.minor.y = element_blank(),
  axis.text.x = element_text(angle = 90, hjust=1))
p_filter
```



Testing algae did not provide results.

Culture age variable.

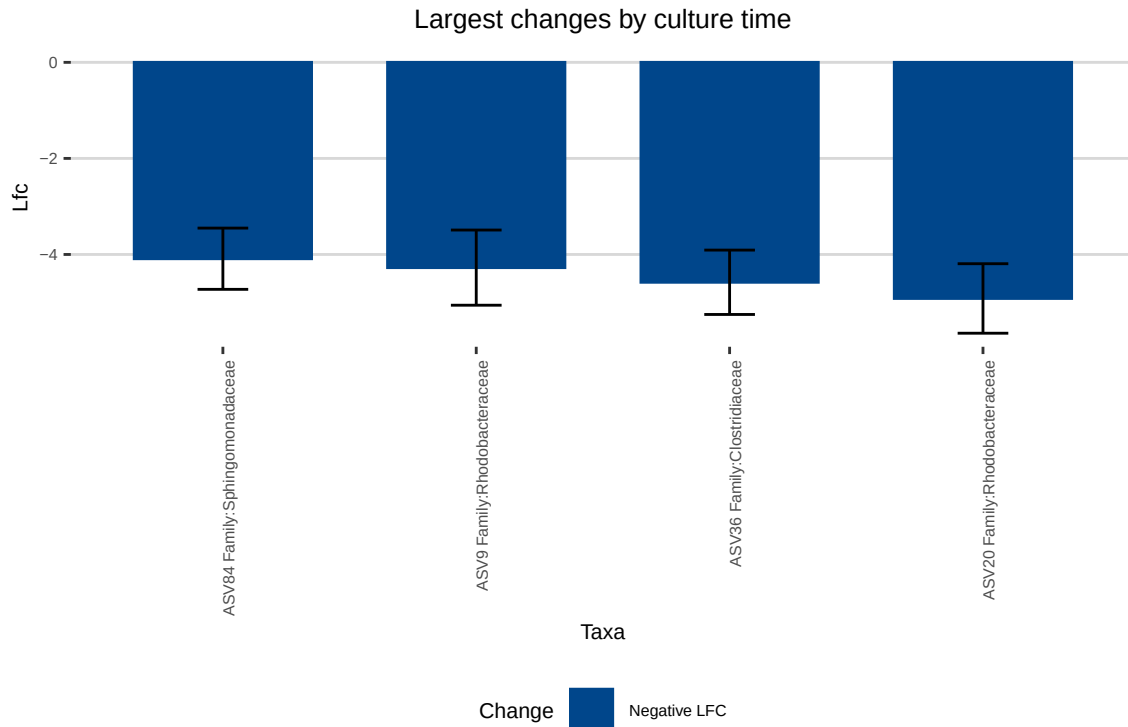
```
#Create new data frame with Lfc, SE, Diff values
df_age <- data.frame(ASV = ancom_out$res$taxon, Lfc = ancom_out$res$lfc_Age5,
  SE = ancom_out$res$se_Age5, Q = ancom_out$res$q_Age5,
  Diff = ancom_out$res$diff_Age5) %>%
  filter(Diff == "TRUE") %>% arrange(desc(Lfc)) %>%
  left_join(featureids, by = "ASV")
df_age$ASV <- paste(df_age$ASV, df_age$taxon)
df_age <- df_age %>% dplyr::select(,-6) %>%
  mutate(Change = ifelse(Lfc > 0, "Positive LFC", "Negative LFC"))
kable(df_age, caption="Taxa that are changed by culture length",
  digits=2,) %>% kable_styling(latex_options = c("HOLD_position",
    "striped"),
  font_size = 12) %>%
  row_spec(0, background="indigo", color="ivory")
```

Table 9: Taxa that are changed by culture length

ASV	Lfc	SE	Q	Diff	Change
ASV84 Family:Sphingomonadaceae	-4.09	0.64	0.04	TRUE	Negative LFC
ASV9 Family:Rhodobacteraceae	-4.27	0.78	0.02	TRUE	Negative LFC
ASV36 Family:Clostridiaceae	-4.58	0.67	0.05	TRUE	Negative LFC
ASV20 Family:Rhodobacteraceae	-4.92	0.72	0.01	TRUE	Negative LFC

Bar plot of LFC including standard error.

```
p_day <- ggplot(data = df_age,
  aes(x = factor(ASV, level=df_age$ASV), y = Lfc,
    fill = Change, color = Change)) +
  geom_bar(stat = "identity", width = 0.7,
    position = position_dodge(width = 0.4)) +
  geom_errorbar(aes(ymin = Lfc - SE, ymax = Lfc + SE), width = 0.2,
    position = position_dodge(0.05), color = "black") +
  labs(x = "Taxa", title = "Largest changes by culture time") +
  theme_hc(base_size=8) + scale_fill_lancet() + scale_color_lancet() +
  theme(plot.title = element_text(hjust = 0.5),
    panel.grid.minor.y = element_blank(),
    axis.text.x = element_text(angle = 90, hjust = 1))
p_day
```



MaAsLin2

```
#Maaslin requires data frame as metadata input
meta_data <- data.frame(colData(tse_daa))
#Counts table needs to be transposed
variant_table <- t(assay(tse_daa))
#Maaslin settings
maaslin_filtering <- Maaslin2(
  variant_table,
  meta_data,
  output = "Maaslin2-filtering",
  transform = "AST",
  fixed_effects = c("Filtered"),
  reference = c("Filtered", "no"),
  normalization = "TSS",
  standardize = FALSE,
  min_prevalence = 0
)
saveRDS(maaslin_filtering, "rds/maaslin_filtering.rds")
```

Note: Maaslin2 will also write results to defined output folder. If you use several fixed effects, it creates also heatmap presentation.

```
maaslin_filtering <- readRDS("rds/maaslin_filtering.rds")
```

Filtering significant results to table by qval value (≤ 0.05).

```
maaslin_table <- maaslin_filtering$results %>%
  dplyr::select(ASV = feature, Coef = coef, SE = stderr, qval,
    N, Nonzero = N.not.zero) %>% filter(qval <= 0.05) %>%
  arrange(desc(Coef))
kable(maaslin_table, digits=2) %>% kable_styling(latex_options = c("HOLD_position",
  "striped"),
  font_size = 10) %>%
  row_spec(0, background="indigo", color="ivory")
```

ASV	Coef	SE	qval	N	Nonzero
ASV3	1.03	0.14	0.00	16	16
ASV382	-0.01	0.00	0.03	16	6
ASV464	-0.01	0.00	0.03	16	6
ASV378	-0.02	0.00	0.03	16	6
ASV98	-0.02	0.00	0.02	16	7
ASV194	-0.02	0.00	0.01	16	7
ASV245	-0.02	0.00	0.00	16	9
ASV249	-0.02	0.01	0.03	16	6
ASV307	-0.02	0.00	0.01	16	7
ASV242	-0.03	0.01	0.03	16	6
ASV239	-0.03	0.01	0.03	16	6
ASV262	-0.03	0.00	0.00	16	8
ASV184	-0.03	0.01	0.03	16	6
ASV193	-0.03	0.01	0.03	16	6
ASV52	-0.03	0.01	0.03	16	10
ASV183	-0.03	0.01	0.03	16	9
ASV85	-0.04	0.01	0.01	16	7
ASV101	-0.04	0.01	0.03	16	7
ASV87	-0.04	0.01	0.01	16	11
ASV119	-0.04	0.01	0.03	16	7
ASV106	-0.05	0.01	0.03	16	8
ASV79	-0.06	0.01	0.03	16	7
ASV78	-0.06	0.01	0.01	16	10
ASV70	-0.06	0.01	0.03	16	8
ASV46	-0.06	0.01	0.03	16	10
ASV64	-0.06	0.01	0.02	16	8
ASV29	-0.09	0.01	0.00	16	12
ASV47	-0.09	0.01	0.00	16	10

Testing algae and age did not provide results.

We can summarize results from different DAA methods and look for common features.

```
#Create daa summaries
aldex_summary <- aldex_res %>% dplyr::select(ASV = genus, Aldex2 = wi.eBH)
ancom_summary <- ancom_out$res %>% dplyr::select(ASV=taxon, Ancombc2 = q_Filteredyes) %>%
  filter(Ancombc2 <= 0.05)
maaslin_summary <- maaslin_filtering$results %>% dplyr::select(ASV=feature, Maaslin2=qval) %>%
  dplyr::filter(Maaslin2 <= 0.05)
#Join three summaries together
daa_summary <- full_join(aldex_summary, ancom_summary, by="ASV")
daa_summary <- full_join(daa_summary, maaslin_summary, by="ASV")
#Create TRUE-FALSE data frame and calculate rowsum score
daa_summary <- daa_summary %>% dplyr::mutate(
  dplyr::across(c(Aldex2:Maaslin2), ~ .x <= 0.05),
  across(-ASV, function(x) ifelse(is.na(x), FALSE, x)),
  Score = rowSums(across(c(Aldex2:Maaslin2)))
) %>% filter(Score > 1)
daa_summary <- daa_summary %>% left_join(featureids, by = "ASV") %>% arrange(ASV)
daa_summary <- daa_summary[c("ASV", "taxon", "Aldex2", "Ancombc2", "Maaslin2", "Score")]
kable(daa_summary, caption="Differential taxa with minimum score of 2") %>%
  kable_styling(latex_options = c("HOLD_position", "striped"),
    font_size = 12) %>%
  row_spec(0, background="indigo", color="ivory")
```

Table 11: Differential taxa with minimum score of 2

ASV	taxon	Aldex2	Ancombc2	Maaslin2	Score
ASV119	Family:Pseudomonadaceae	FALSE	TRUE	TRUE	2
ASV29	Family:Carnobacteriaceae	TRUE	TRUE	TRUE	3
ASV3	Genus:Pseudomonas	FALSE	TRUE	TRUE	2
ASV64	Genus:Brevundimonas	TRUE	FALSE	TRUE	2
ASV70	Genus:Dyadobacter	TRUE	FALSE	TRUE	2
ASV78	Family:Comamonadaceae	TRUE	FALSE	TRUE	2